

AI-Powered Skill Classification: Mapping Technology Intensity in the German Labor Market

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EALE Conference 2026

Published in *Fiscal Studies* (2026)

<https://onlinelibrary.wiley.com/doi/10.1111/1475-5890.70020>

Motivation

- ▶ Rapid technological change is reshaping skill demands and transforming entire occupations.
- ▶ Digital technologies (computers, industrial robots) have long been integrated into jobs.
- ▶ **Frontier technologies** – above all **artificial intelligence** – are the key drivers of recent change.

Open question

How do frontier technology skills spread across jobs? Measurement is hard: these technologies are complex, fast-moving, and rarely named explicitly in occupational data.

- ▶ Existing measures are mostly *forward-looking exposure* indices (Frey/Osborne 2017; Felten et al. 2018; Webb 2020; Brynjolfsson et al. 2018) – they capture what technology *could* do, not what skills jobs *actually* require.

This paper

Contribution

A new **skill-based measure of occupational technology intensity** that distinguishes **manual**, **digital**, and **frontier** technology skills – capturing the technology content *actually implemented* in jobs.

What we do

- ▶ Classify 8,775 job skills from the German expert database **BERUFENET** using NLP, generative AI (GPT-4o), and supervised machine learning – an *AI-powered* but transparent and reproducible classification.
- ▶ Aggregate to occupations and link to **administrative employment data** (30.4 million workers), 2012–2023.

Main findings

- ▶ Average worker in 2023: **42% manual**, **38% digital**, **20% frontier** skill content.
- ▶ Broad shift from manual and digital toward frontier skills within nearly all occupations.
- ▶ **U-shaped** relationship between changes in frontier skill intensity and employment growth.

Concept: three technology levels of skills

SKILL LEVEL	SERVICE	PRODUCTION
Frontier technologies performing work processes <i>autonomously</i>	IT-integrated machine learning, fraud detection, chatbots	Self-controlled collaborative robots, digital twins, autonomous driving
Digital humans <i>indirectly</i> involved in work processes	IT-supported CRM software, reservation systems, ERP	Indirectly controlled CNC, fire alarm systems, CAD
Manual humans conduct work processes <i>manually</i>	Not IT-supported voice teaching, group dancing, social therapy	Manually controlled rust removal, tile laying, metalwork

Framework builds on Genz et al. (2021). “Manual” is a residual category that also absorbs non-technological (social, managerial, creative) skills.

Data

BERUFENET expert database (German Federal Employment Agency)

- ▶ 8,775 unique job skills linked to 4,647 occupations (8-digit KldB2010) in 2012/2023.
- ▶ Regularly updated by experts: reflects skills *currently required* in each occupation.
- ▶ Only short skill *titles* – no standardized definitions.

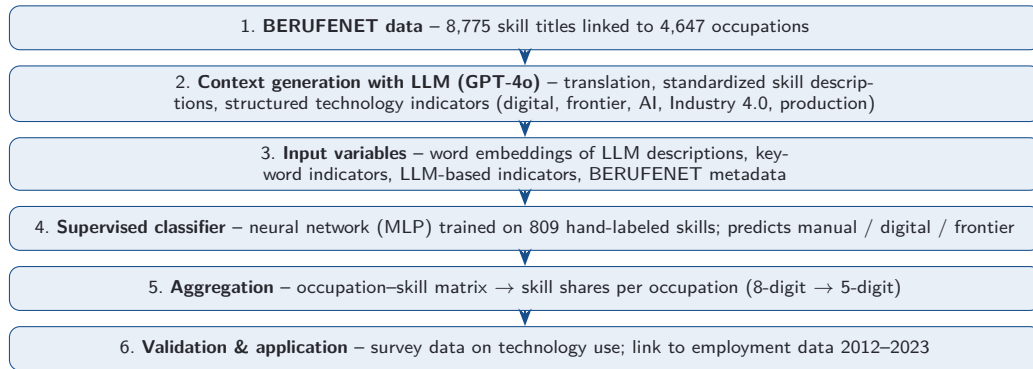
Example: *data scientist* (36 skills in 2023)

“applied computer science”, “data lake”, “machine learning”, “programming”, “project management”, “IT coordination”, ...

Administrative employment data (IAB Employment History, BeH)

- ▶ All employees subject to social insurance, aged 16–65: **30.4 million individuals** in 2023.
- ▶ Occupations at the 5-digit KldB2010 level (1,297 occupations), 2012–2023.

Method: AI-powered skill classification pipeline



The LLM enriches skill titles with semantic context – the classification itself is done by the supervised model: transparent and reproducible. **example: GenAI output**

Does the AI-powered approach pay off?

	Accuracy	Cohen's κ	Recall (Digital)	Recall (Frontier)
<i>Rule-based text-mining</i>				
on skill titles	0.637	0.434	0.420	0.418
on LLM-generated descriptions	0.760	0.639	0.871	0.552
<i>AI-powered classifier (MLP)</i>				
title embeddings only	0.860	0.790	0.837	0.866
descriptions + structured features	0.932	0.897	0.920	0.895

- ▶ LLM-generated semantic context is essential for identifying **frontier** technologies that are not explicitly named in skill titles (e.g., *Microsoft Azure*, specific SAP modules).
- ▶ Full classifier: macro-averaged F1 close to 0.8 on a held-out test set.

feature importance

skill examples

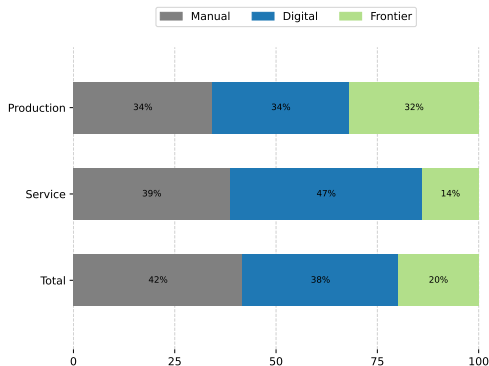
The technology profile of the German labor market in 2023

Occupational Technology Skill Share (OTSS) for occupation j and technology class i :

$$\text{OTSS}_{ij} = \frac{\# \text{ skills of class } i \text{ in } j}{\# \text{ skills in } j}$$

with $i \in \{\text{manual}, \text{digital}, \text{frontier}\}$.

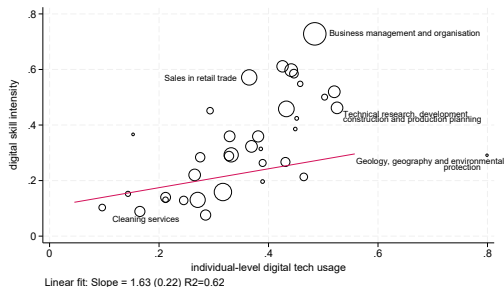
Constructed at the 8-digit level, averaged to 5-digit occupations, aggregated with employment weights.



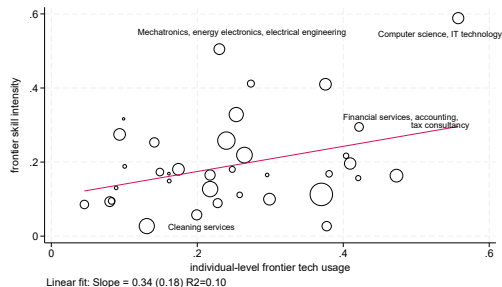
- ▶ Average worker: **42% manual**, **38% digital**, **20% frontier** skill content.
- ▶ Production work is more frontier-intensive (32%) than services (14%); services are dominated by digital skills (47%).

Validation: survey data on actual technology use

(a) **Digital** skills



(b) **Frontier** skills



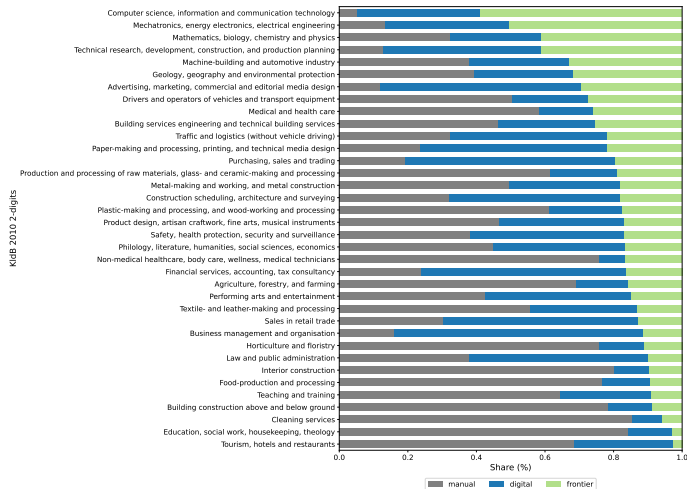
- ▶ DiWaBe 2.0 survey ($\approx 9,800$ employees): self-reported workplace technology use.
- ▶ Strong correlation for **digital** skills (0.786, $p < 0.01$); positive for **frontier** skills (0.309, $p < 0.10$) $\Rightarrow$ our measures capture *actual* technology use.

Who works in frontier vs. digital occupations?

Occupation (KldB2010, 5-digit)	# employees	Man.	Dig.	Front.
<i>Most common high-frontier-skill occupations</i>				
Technical R&D occupations (highly complex)	218,216	8	34	58
Software developers (highly complex)	203,338	5	31	64
Construction electricians (skilled)	177,929	12	35	53
IT application consultants (highly complex)	114,505	9	33	58
ICT specialists (skilled)	103,412	9	38	54
<i>Most common high-digital-skill occupations</i>				
Office clerks and secretaries (skilled)	1,487,732	11	82	7
Business administration (skilled)	771,916	14	80	6
Accountants (complex)	230,517	24	69	8
Managing directors & executive board	216,197	16	65	19
Office clerks and secretaries (un-/semi-skilled)	213,301	10	81	9

- **Frontier** occupations: concentrated in technical, engineering, IT domains – skilled and complex jobs.
- **Digital** occupations: much broader domains and *far larger* employment.

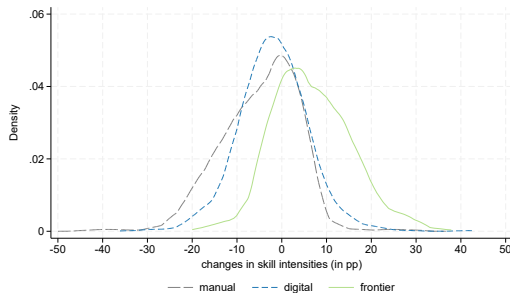
Frontier skills are concentrated – digital skills are everywhere



- ▶ Top: computer science / ICT (**59% frontier**), mechatronics & electrical engineering (**50%**).
- ▶ Bottom: cleaning services (**85% manual**).

Employment-weighted skill shares by occupational group (2-digit KldB2010), ranked by frontier share. **by industry**

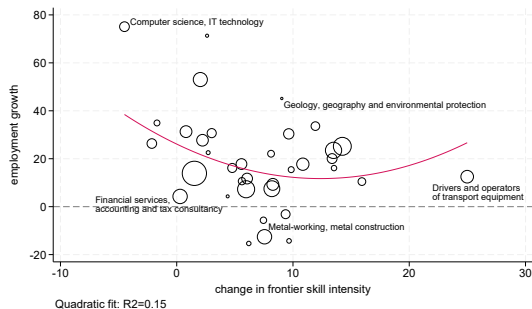
Within-occupation change, 2012–2023: the rise of frontier skills



- ▶ Within 5-digit occupations: **manual** -4.8pp , **digital** -1.9pp , **frontier** $+6.7\text{pp}$ on average.
- ▶ Frontier shares rose in 914 of 1,181 occupations existing in both years – a broad-based diffusion, not a niche phenomenon.

levels 2012 vs. 2023

Frontier skill change and employment growth: a U-shape



- ▶ Employment growth is strongest at the *extremes*: occupations with very small or very large increases in frontier skill intensity; weakest for intermediate changes.
- ▶ Pattern holds across age, gender, and qualification groups – complementarity for some occupations, substitution pressure for others.

Open data: the OTSS repository and dashboard

- ▶ Occupational technology skill shares (**manual** / **digital** / **frontier**) freely available for download and reuse.
- ▶ All intensities **interactively** explorable in the dashboard: filter by occupation, occ. group, and technology level.
- ▶ For researchers, policymakers, career guidance, and teaching.

OTSS – Occupational Technology Skill Shares

Data repository:

<https://github.com/otss-data/otss>

Dashboard:

<https://otss-data.github.io/otss-dashboard>



Scan to open the dashboard

Takeaways

1. **New measure:** AI-powered, skill-based technology intensity of occupations – transparent, reproducible, validated against survey data on actual technology use.
2. **2023 snapshot:** manual 42% > digital 38% > frontier 20%; frontier skills concentrated in technical/IT occupations, digital skills pervasive.
3. **2012–2023:** broad within-occupation shift from manual and digital toward frontier skills.
4. **Employment:** U-shaped relation between frontier skill changes and employment growth – polarization, not uniform gains.

Beyond this paper

The method travels: applicable to O*NET and other occupational databases, to job postings, and to other skill domains (e.g., “green” skills) – a near-real-time monitor of technological transformation.

Thank you!

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Genz, Gregory & Lehmer (2026), *Fiscal Studies*
<https://onlinelibrary.wiley.com/doi/10.1111/1475-5890.70020>

Data & dashboard: <https://github.com/otss-data/otss>

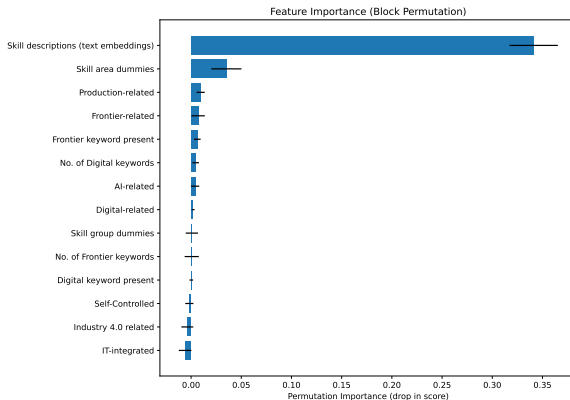
Backup: GenAI-generated skill descriptions and technology indicators

Skill title (BERUFENET)	Skill description generated with GenAI (abridged)	GenAI-generated indicators						
		Dig.	Front.	AI	I4.0	IT-int.	Self-c.	Prod.
Speechwriting	Crafting speeches that communicate a message to an audience; organizing ideas, persuasive language, rhetorical devices to engage listeners.	0	0	0	0	0	0	0
Project planning software MS Project	Digital project-management tool to plan, schedule and manage projects: task assignment, resource allocation, Gantt charts; integrates with MS Office.	1	0	0	0	1	0	0
Machine learning	Subset of artificial intelligence using algorithms and statistical models so that computers improve through experience and data; applications include natural language processing and image recognition.	1	1	1	1	1	0	0
Autonomous driving systems	Enables vehicles to navigate without human intervention, integrating sensors, cameras, radar and LiDAR with machine-learning models; lane keeping, adaptive cruise control, traffic sign recognition.	1	1	1	1	0	1	1

Small version of paper Table 3, descriptions abridged. Indicators: digital, frontier, AI-related, Industry 4.0, IT-integrated, self-controlled, production-related. Descriptions and indicators enter the supervised classifier as input features.

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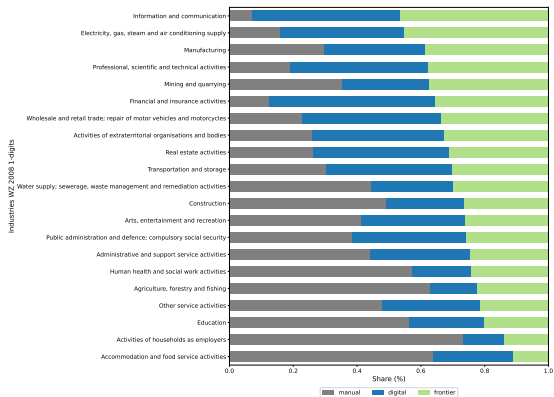
Backup: feature importance of the classifier



Permutation-based feature importance. Embeddings of LLM-generated skill descriptions dominate; structured indicators and metadata play a secondary role.

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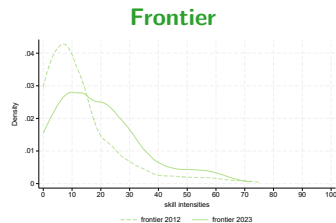
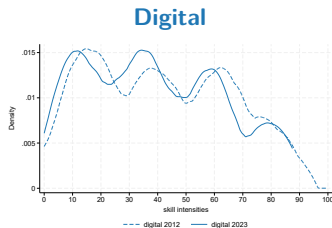
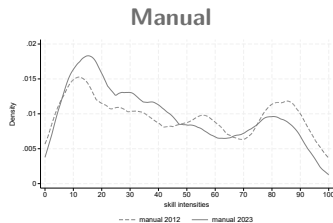
Backup: ranking by industry



Employment-weighted skill shares by industry (WZ 2008, 1-digit).

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Backup: distribution of skill intensity levels, 2012 vs. 2023



Employment-weighted kernel densities of occupational skill intensities (5-digit KldB2010). Frontier intensities shift rightward across the entire distribution.

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Backup: examples of classified skills

- ▶ **Frontier:** machine learning, augmented reality, data lake, maintenance and repair robots, computer-assisted surgery, networked production systems, LiDAR, Microsoft Azure cloud computing.
- ▶ **Digital:** bookkeeping, CNC programming, office management, CRM software, e-mail programs, web content management systems.
- ▶ **Manual:** sealing (roof), pipe construction, wood sorting, infant and toddler care, art education, social therapy.

Occupation-level examples

Data scientist: 72% frontier, 28% digital, 0% manual.

Autonomous driving engineer: 88% frontier, 12% digital.

Customer information specialist: 8% frontier, 92% digital.

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